Integral Calculus 1

Introduction

Integral Calculus is the branch of calculus which deals with functions to be integrated. **Integration** is the reverse process of differentiation. The function to be integrated is referred to as the **integrand** while the result of an integration is called **integral**.

The integral symbol or sign ∫, is an elongated S denoting sum (Latin: *summa*), was introduced by Leibniz, who named integral calculus as ***calculus summatorius***.

Definite & Indefinite Integrals

Definite integral is an integral that is defined by the limit values a and b of the independent variable.

Example: $\int_a^b f(x)dx$

Indefinite integral is an integral with no restrictions imposed on its independent variables. It is also called antiderivative or primitive integral.

Example: $\int f(x)dx$

Fundamental Theorem of Calculus

Suppose f is a continuous function in a closed interval [a, b], then:

$F'(x) = f(x)$ for all x in [a, b].

The fundamental theorem of calculus states that the definite integral is:

$$\int_a^b f(x)dx = F(b) - F(a)$$

Basic Integrals

1. $\int du = u + C$
2. $\int a\,du = au + C$
3. $\int u^n du = \frac{u^{n+1}}{n+1} + C \quad (n \neq -1)$
4. $\int \frac{du}{u} = \ln u + C$

Exponential & Logarithmic Functions

5. $\int e^u du = e^u + C$

6. $\int a^u du = \frac{a^u}{\ln a} + C$
7. $\int ue^u du = e^u(u-1) + C$
8. $\int \ln u\,du = u\ln u - u + C$
9. $\int \frac{du}{u\ln u} = \ln|\ln u| + C$

Trigonometric Functions

10. $\int \sin u\,du = -\cos u + C$
11. $\int \cos u\,du = \sin u + C$
12. $\int \tan u\,du = \ln|\sec u| + C$
13. $\int \cot u\,du = \ln|\sin u| + C$
14. $\int \sec u\,du = \ln|\sec u + \tan u| + C$
15. $\int \csc u\,du = \ln|\csc u - \cot u| + C$
16. $\int \sec^2 u\,du = \tan u + C$
17. $\int \csc^2 u\,du = -\cot u + C$
18. $\int \sec u \tan u\,du = \sec u + C$
19. $\int \csc u \cot u\,du = -\csc u + C$
20. $\int \sin^2 u\,du = \frac{1}{2}u - \frac{1}{4}\sin 2u + C$
21. $\int \cos^2 u\,du = \frac{1}{2}u + \frac{1}{4}\sin 2u + C$
22. $\int \tan^2 u\,du = \tan u - u + C$
23. $\int \cot^2 u\,du = -\cot u - u + C$

Inverse Trigonometric Functions

24. $\int \sin^{-1} u\,du = u\sin^{-1} u + \sqrt{1-u^2} + C$
25. $\int \cos^{-1} u\,du = u\cos^{-1} u - \sqrt{1-u^2} + C$
26. $\int \tan^{-1} u\,du = u\tan^{-1} u - \ln\sqrt{1-u^2} + C$
27. $\int \cot^{-1} u\,du = u\cot^{-1} u + \ln\sqrt{1+u^2} + C$
28. $\int \sec^{-1} u\,du = \left[\begin{array}{l} u\sec^{-1} u \\ -\ln\left|u + \sqrt{u^2-1}\right| + C \end{array}\right]$
29. $\int \csc^{-1} u\,du = \left[\begin{array}{l} u\csc^{-1} u \\ +\ln\left|u + \sqrt{u^2-1}\right| + C \end{array}\right]$

Hyperbolic Functions

30. $\int \sinh u\,du = \cosh u + C$
31. $\int \cosh u\,du = \sinh u + C$
32. $\int \tanh u\,du = \ln|\cosh u| + C$
33. $\int \coth u\,du = \ln|\sinh u| + C$
34. $\int \operatorname{sech} u\,du = \tan^{-1}(\sinh u) + C$
35. $\int \operatorname{csch} u\,du = \ln\left|\tanh \frac{u}{2}\right| + C$
36. $\int \operatorname{sech}^2 u\,du = \tanh u + C$
37. $\int \operatorname{csch}^2 u\,du = -\coth u + C$
38. $\int \operatorname{sech} u \tanh u\,du = -\operatorname{sech} u + C$
39. $\int \operatorname{csch} u \coth u\,du = -\operatorname{csch} u + C$
40. $\int \sinh^2 u\,du = \frac{1}{4}\sinh 2u - \frac{1}{2}u + C$
41. $\int \cosh^2 u\,du = \frac{1}{4}\sinh 2u + \frac{1}{2}u + C$